

Name: _____

Teacher: _____



MATHEMATICS

4 UNIT PAPER

HSC
Assessment N° 1

December 2007

Time Allowed - **70 minutes**

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- Questions are of different value.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.

The question paper must be handed to the supervisor at the end of the examination.

- 1 If $(1+i)z = 3-i$, find
- a) z in the form $x+iy$ [2]
 - b) $|z|$ [1]
 - c) $\arg z$ [1]
 - d) $\bar{z}$ [1]
- 2 If z is a complex number prove that $|z|^2 = z \times \bar{z}$ [3]
- 3 If the real part of $\frac{z-4}{z-2i}$ is zero, prove that the locus of the point representing z in the Argand Diagram is a circle of radius $\sqrt{5}$ units. [5]
- 4 sketch the region of the Argand Diagram whose points z satisfy the inequalities
- a) $|z-\bar{z}| \leq 4$ and $-\frac{\pi}{3} \leq \arg z \leq \frac{\pi}{3}$. [3]
 - b) $1 < \operatorname{Re}(z) < 2$ and $\arg\left(\frac{z+1}{z-1}\right) \leq \frac{\pi}{4}$. [4]
- 5 a) Write the complex number $1-\sqrt{3}i$ in modulus-argument form. [2]
- b) Hence use De Moivre's theorem to find $(1-\sqrt{3}i)^{10}$ in the form $a+ib$, where a, b are real numbers. [2]

- 6 a) If ω is a complex cube root of unity, show that ω^2 is the other complex cube root. [2]
- b) Evaluate $\frac{\omega^2}{2} - \frac{\omega + 1}{2\omega}$ [2]
- 7 The complex number z is given by $z = -1 - i\sqrt{3}$
- a) Write the values of $\arg z$ and $|z|$. [2]
- b) If n is integral, explain why z^n is never purely imaginary. [3]
- 8 a) Find the six sixth roots of -1 . Leave your answers in modulus-argument form. [4]
- b) Hence, or otherwise solve $z^4 - z^2 + 1 = 0$. [3]

[total 40 marks]

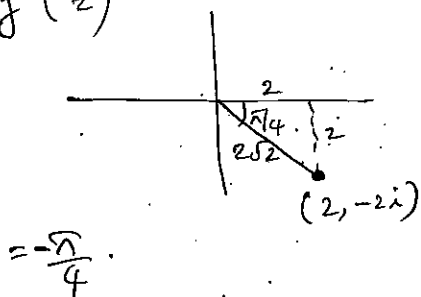


$$(1+i)z = 3-i$$

$$\begin{aligned} a) \quad z &= \frac{3-i}{1+i} \\ &= \frac{(3-i)(1-i)}{(1+i)(1-i)} \\ &= \frac{3-3i-i+1}{2} \\ &= \frac{4-4i}{2} = 2-2i \end{aligned}$$

$$\begin{aligned} (b) \quad |z| &= \sqrt{2^2 + (-2)^2} \\ &= 2\sqrt{2} \end{aligned}$$

$$(c) \quad \arg(z)$$



$$(1) \quad \bar{z} = \overline{2-2i} = 2+2i$$

$$2. \text{ let } z = x+iy$$

$$|z|^2 = x^2 + y^2 \quad \text{--- (1)}$$

$$\begin{aligned} z\bar{z} &= (x+iy)(x-iy) \\ &= x^2 + y^2 \quad \text{--- (2)} \end{aligned}$$

Using (1) & (2),

$$|z|^2 = z\bar{z}$$

$$3. \quad \frac{z-4}{z-2i} = \frac{x+iy-4}{x+iy-2i}$$

$$= \frac{x-4+iy}{x+i(y-2)}$$

$$= \frac{(x-4+iy)[x-i(y-2)]}{[x+i(y-2)][x-i(y-2)]}$$

$$= \frac{x^2-4x+xyi-xyi-4iy+8i}{x^2+(y-2)^2}$$

$$= \frac{x(x-4)+i(x-4)(y-2)+xyi+y^2-2y}{x^2+(y-2)^2}$$

$$= \frac{x^2-4x+xyi-2xi-4yi+8i+xyi+y^2-2y}{x^2+(y-2)^2}$$

$$= \frac{x^2-4x+y^2-2y+i(2xy-2x-4y+8)}{x^2+(y-2)^2}$$

$$\operatorname{Re}\left(\frac{z-4}{z-2i}\right) = 0$$

$$i.e. \frac{x^2-4x+y^2-2y}{x^2+(y-2)^2} = 0$$

$$\therefore x^2-4x+y^2-2y=0$$

$$i.e. x^2-4x+4+y^2-2y+1=5$$

$$(x-2)^2+(y-1)^2=5$$

Circle with centre (2,1) and
radius $\sqrt{5}$.

④ a) $|z - \bar{z}| \leq 4$

and $-\frac{\pi}{3} \leq \arg(z) \leq \frac{\pi}{3}$.

$z = x + iy$.

$|z - \bar{z}| = |x + iy - x + iy|$

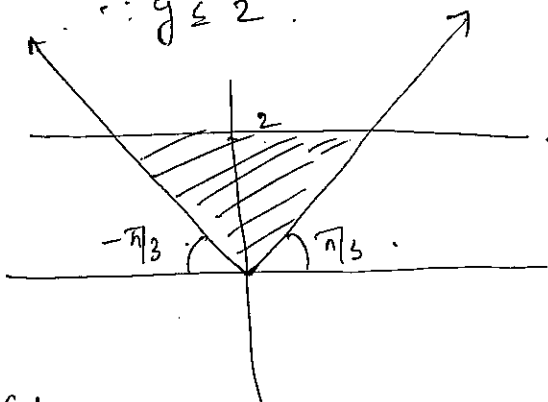
$= |2iy| \leq 4$.

$|0 + 2iy| \leq 4$.

$\sqrt{0 + (2y)^2} \leq 4$.

$2y \leq 4$

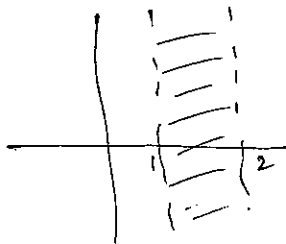
$\therefore y \leq 2$.



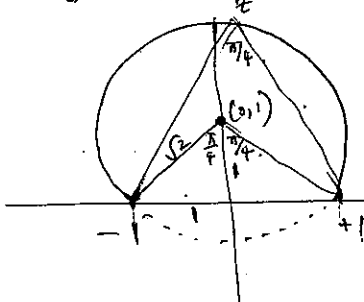
(b) $1 < \operatorname{Re}(z) < 2$ and $\arg\left(\frac{z+1}{z-1}\right) \leq \pi/4$.

$\operatorname{Re}(z) = x$.

$1 < x < 2$



$\arg\left(\frac{z+1}{z-1}\right) \leq \pi/4$.

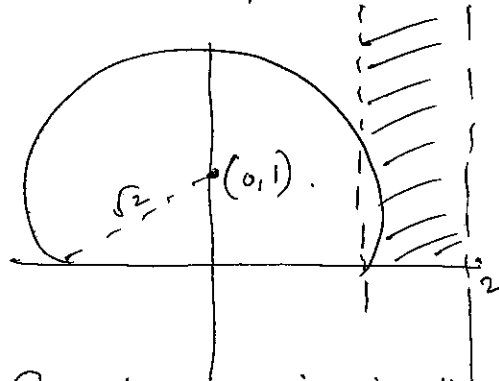


If z was outside the circle, the angle will be less than $\pi/4$.

$\therefore$ equation of this circle is

$x^2 + (y-1)^2 = 2$.

when $\arg\left(\frac{z+1}{z-1}\right) \leq \pi/4$, it is the outside of this circle.

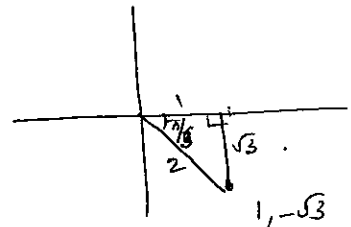


The shaded region is the required area.

⑤ $1 - \sqrt{3}i$

a)

$= 2 \cos\left(\frac{\pi}{3}\right)$



(b) $(1 - \sqrt{3}i)^{10}$
 $= \left(2 \cos\left(\frac{\pi}{3}\right)\right)^{10} = 2^{10} \cos\left(\frac{10\pi}{3}\right)$
 $= 1024 \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)$
 $= 512 (-1 + \sqrt{3}i)$
 $= -512 (1 - \sqrt{3}i)$

6. (a) Consider the equation $z^3 = 1$.

Let $z = r \operatorname{cis} \theta$.

$$r^3 \operatorname{cis} 3\theta = 1. \quad r = 1.$$

$$\therefore \cos 3\theta = 1; \sin 3\theta = 0$$

$$\therefore 3\theta = 2k\pi + 0.$$

$$\therefore \theta = \frac{2k\pi}{3} \quad k = 0, 1, 2 \dots$$

$$k = 0. \quad z_1 = \operatorname{cis} 0 = 1.$$

$$k = 1. \quad z_2 = \operatorname{cis} \frac{2\pi}{3} = \omega.$$

$$k = 2. \quad z_3 = \operatorname{cis} \frac{4\pi}{3} = \left(\operatorname{cis} \frac{2\pi}{3}\right)^2 = \omega^2$$

(using De-Moivre theorem)

$\therefore 1, \omega, \omega^2$ are the roots of unity.

$$(b) \quad \frac{\omega^2}{2} - \frac{\omega+1}{2\omega}$$

$$= \frac{1}{2} \left(\omega^2 - \frac{\omega+1}{\omega} \right)$$

$$= \frac{1}{2} \left(\frac{\omega^3 - \omega - 1}{\omega} \right)$$

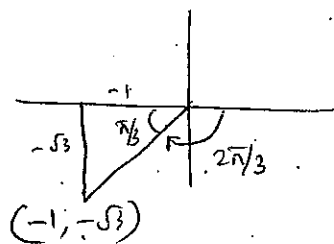
$$= \frac{1}{2} \left(\frac{1 - \omega - 1}{\omega} \right)$$

(ω^3 being $= 1$)

$$= \frac{1}{2} \times -1 = -\frac{1}{2}$$

$$(7) \quad z = -1 - i\sqrt{3}$$

(a)



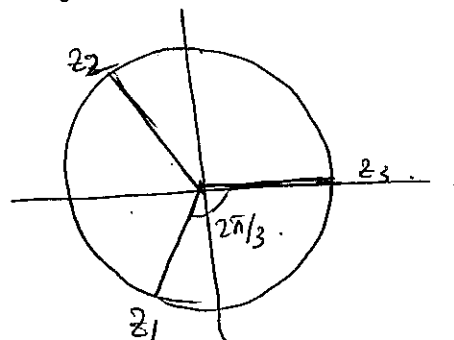
$$\arg(z) = -\frac{2\pi}{3}$$

$$|z| = \sqrt{(-1)^2 + (-\sqrt{3})^2} = 2$$

$$(b) \quad z = 2 \operatorname{cis} \left(-\frac{2\pi}{3}\right)$$

$$\therefore z^n = 2^n \operatorname{cis} \left(-\frac{2n\pi}{3}\right)$$

On an Argand diagram the roots of $z^n = 1$ are as given below for various integral values of n .



This shows that the z^n is either complex or real.

i.e. for no integral value of n , z^n is purely imaginary.

$$(8)(a) \quad z^6 = -1.$$

$$\text{Let } z = r \operatorname{cis} \theta, \quad r = 1.$$

$$\therefore z = \operatorname{cis} \theta.$$

$$z^6 = \operatorname{cis} 6\theta = -1.$$

$$\therefore \cos 6\theta = -1 \quad \sin 6\theta = 0.$$

$$6\theta = 2k\pi + \pi \\ = (2k+1)\pi$$

$$\therefore \theta = \frac{(2k+1)\pi}{6}, \quad k=0,1,2,3,4,5.$$

$$k=0, \quad z_1 = \operatorname{cis} \frac{\pi}{6} + i \sin \frac{\pi}{6}$$

$$k=1, \quad z_2 = \operatorname{cis} \frac{\pi}{2} = i$$

$$k=2 \quad z_3 = \operatorname{cis} \frac{5\pi}{6}.$$

$$\theta = (2k+1)\frac{\pi}{6}, \quad k=0, \pm 1, \pm 2, -3.$$

$$k=0 \quad z_1 = \operatorname{cis} \frac{\pi}{6}.$$

$$k=1 \quad z_2 = \operatorname{cis} \frac{\pi}{2} = i$$

$$k=-1, \quad z_3 = \operatorname{cis} \left(-\frac{\pi}{6}\right).$$

$$k=-2 \quad z_4 = \operatorname{cis} \left(-\frac{\pi}{2}\right) = -i$$

$$k=2 \quad z_5 = \operatorname{cis} \left(\frac{5\pi}{6}\right).$$

$$k=-3 \quad z_6 = \operatorname{cis} \left(-\frac{5\pi}{6}\right).$$

$$(b) \quad z^6 + 1 = (z^2)^3 + 1$$

$$= (z^2 + 1)(z^4 - z^2 + 1).$$

$$z^2 + 1 \text{ have roots } z = \pm i.$$

$$\therefore \text{Roots of } z^4 - z^2 + 1 = 0 \text{ are}$$

$$\operatorname{cis} \left(\pm \frac{\pi}{6}\right), \operatorname{cis} \left(\pm \frac{5\pi}{6}\right).$$

$$= \underline{\underline{\frac{\sqrt{3}}{2} \pm \frac{1}{2}i, \quad -\frac{\sqrt{3}}{2} \pm \frac{1}{2}i}}$$